

CS5013 Project Proposal

Exchange Student Guide & Community Platform

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1 Team information

Team ID: team-EXCHANGE.

- Member 1: Mikhail Novikov – roll no. GE26Z858 – department CSE – GitHub handle rikire
- Member 2: Abdirakhim Ismailov – roll no. GE26Z860 – department CSE – GitHub handle abdra04-gif

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1.1 Module ownership (preliminary)

- **Backend (Mikhail)**
- **Frontend (Abdirakhim)**
- **Shared:** DevOps, documentation, stakeholder communication.

2 Problem statement

The main problem for incoming exchange students at IIT Madras is the overwhelming amount of scattered information. Data about hostel check-in, SIM cards, bank cards, and food options are spread across different places, and in this flood of information it is easy to miss something important or come across incomplete details. There is no single convenient source where all the necessary information is gathered in one place. We will build a centralised knowledge base where all information is structured and collected in one place, so that every new student can quickly find everything they need and not miss anything important.

3 Users and stakeholder

Named stakeholder: Mr. Thukaram, Lead, Inbound International Academic Programs, Office of Global Engagement (OGE), IIT Madras.

We reached out to Mr. Thukaram on August 14, 2026, with the project idea. He agreed to be our stakeholder and confirmed that a centralised knowledge base would be useful for his office and for incoming exchange students. He will provide feedback on content categories and test the final tool. His acknowledgement email is attached as Appendix A.

4 Prior work and why it does not suffice

1. **OGE Orientation Presentations** – internal materials (PDF/PPT). Overlap: Provide basic information about IIT Madras and initial procedures. Gap: The information is superficial and insufficient for students' practical needs.
2. **IIT Madras Official Website** – <https://www.iitm.ac.in/>. Overlap: Contains official information about accommodation, fees, and academic regulations. Gap: There is a lot of information, but it is scattered across different pages and sections. Finding what you need is difficult and time-consuming. In addition, the information is often outdated and not regularly updated.
3. **WhatsApp Groups for International Students** – internal chats. Overlap: Used for asking and answering questions. Gap: Information is unstructured and gets lost in hundreds of messages. Finding old answers is difficult. The same questions are asked repeatedly, and there is no search function.
4. **Travel Blogs and City Websites** – external (e.g., TripAdvisor). Overlap: Provide recommendations on places to visit and eat in Chennai. Gap: They are not specific to IIT Madras. They do not describe places within the campus and often contain inaccurate or outdated information about where to go and what to see.

5 Tech stack

Language: **Java** (course policy).

- Build system: Maven
- Backend Framework: Spring Boot (REST API) – owned by Mikhail
- Frontend: Thymeleaf with Bootstrap 5 – owned by Abdirakhim
- Database: H2 (development) / PostgreSQL (production)
- Search: Hibernate Search with Lucene (full-text search)
- Key libraries: Spring Data JPA, Hibernate Search, Thymeleaf, Lombok, JUnit 5
- Where the tool will run: Containerised using Docker. The application can be deployed on any platform that supports Docker containers (Render.com, Fly.io, AWS, Azure, Google Cloud, or a local server).

6 AI-usage statement

We will use AI coding assistants (such as Claude Code, GitHub Copilot, ChatGPT, or Gemini) during this project, in line with the course policy on AI-assisted programming. Every line of code we submit will be code we understand and can explain in the viva, whether we typed it or the assistant generated it.

7 Verification plan

Test fixtures we will use:

- **Fixture 1:** We will populate the knowledge base with articles on the main topics that concern incoming exchange students. These topics were identified through our communication with OGE: SIM cards, bank cards, hostel check-in, food, prices, useful places. The base will contain 20–30 articles - enough to test search and navigation.

- **Fixture 2:** Synthetic corner-case data: empty search queries, SQL injection attempts, HTML tags in article content, duplicate article submissions, very long titles (100+ characters).
- **Fixture 3:** Performance test: 100 articles with 500 words each to verify search speed (≤ 2 seconds).

User-visible acceptance criteria the stakeholder will use to say “yes, this works”:

- Mr. Thukaram can open the application URL, see the homepage with all 6 categories listed, and click on any category to see all articles.
- He can type “FRRO registration” in the search box and get at least 2 relevant articles within 2 seconds.
- He can submit a new article titled “How to get a library card” via the “Add Article” form, and it appears immediately on the homepage.
- He can edit an existing article (e.g., update the phone number) and the change is visible immediately.

8 Milestone plan

- **By the design doc (11 Sep 2026):** Database schema designed, Spring Boot project scaffolded with JPA entities and CRUD operations, and prototype homepage with static categories ready.
- **By the mid-demo (9 Oct 2026):** Core workflow runs end-to-end: category browsing, full-text search works on 30+ articles, article submission and editing works, admin panel with single password, and both team members contributing to git.
- **By the final submission (6 Nov 2026):** Stakeholder has successfully used all features, video recording of stakeholder using the tool, comprehensive README, unit tests covering 80%+ of business logic, and handoff documentation for non-technical staff.

9 Risks and plan B

1. **Risk:** Thymeleaf + server-side rendering is not interactive enough for some features (dynamic filters, instant updates). **Plan B:** Integrate React (or Alpine.js) for individual components without rewriting the whole project. We will keep Thymeleaf for page structure and add React for complex parts.
2. **Risk:** Hibernate Search with Lucene may not work on Render (ephemeral filesystem) or may be too slow for indexing. **Plan B:** Switch to PostgreSQL Full-Text Search (‘tsvector’/‘tsquery’). This avoids external indexing and works on any platform with PostgreSQL.
3. **Risk:** We do not have access to the IIT Madras servers and cannot deploy the application inside the university network. This may be an issue if the stakeholder expects the tool to run within the university infrastructure. **Plan B:** We will deliver the repository (code, Dockerfile, and build instructions) to Mr. Thukaram. He will forward it to the university IT team, who will deploy it on their servers. This is their standard practice — they are familiar with deploying Docker containers. We will also provide a detailed step-by-step deployment guide with commands and configuration notes.

A Stakeholder acknowledgement email

Dear Mikhail,

Thank you for your email and for sharing the detailed information regarding the

CS5013: Programming with AI course project.

I am happy to support your teams as the stakeholder for the proposed projects. Based on the scope and potential usefulness to the Office of Global Engagement, I would suggest the following:

Team A – Novikov Mikhail and Abdirakhim Ismailov

Option 2: Exchange Student Guide & Community Platform

This would be useful for incoming exchange students by providing a centralised resource covering administrative procedures, campus facilities, essential services, and practical information about life at IIT Madras and Chennai. The proposed MVP scope appears appropriate for the project timeframe.

Team B – Iurii Rudenko

Option 3: Exchange Student Testimonial & Photo Gallery

This would be a focused and manageable project for a solo developer. It could also be useful for showcasing exchange students' experiences and supporting OGE's future international student outreach and engagement activities.

I am happy to act as the stakeholder for both projects and provide the necessary inputs during the development period. We can have a short meeting to finalise the scope, prioritise the features, and clarify the information/resources that can be provided by the Office of Global Engagement.

You may use this email as my preliminary acknowledgement of willingness to act as the stakeholder for the respective project proposals.

All the best for your project, and I look forward to working with you.

Thanks and regards,

Thukaram M Damodhar

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